No. 7 13.11.2024

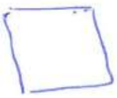
$\boxed{1}$

$$W = \int_\Omega F(u, P_1, P_2)\, d\Omega \quad \rightarrow \quad \frac{\partial F}{\partial u} - \frac{\partial}{\partial x}\frac{\partial F}{\partial P_1} - \frac{\partial}{\partial y}\frac{\partial F}{\partial P_2} = 0$$

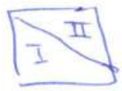
$$P_1 = \frac{\partial u}{\partial x}, \quad P_2 = \frac{\partial u}{\partial y}$$

$$W_h = \sum_{\Omega_h} F(d_0^+ u, d_1^+ u, d_2^+ u)\, \Delta S \rightarrow d_0^- \frac{\partial F}{\partial d_0^+ u} - d_1^- \frac{\partial F}{\partial d_1^+ u} - d_2^- \frac{\partial F}{\partial d_2^+ u} = 0$$

2D:



будет правильно



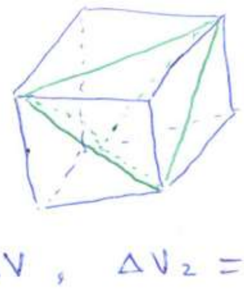
$$\Delta S = \Delta S_{I} + \Delta S_{II} \Rightarrow \Delta S_1 = \Delta S_2 = \frac{1}{2}\Delta S$$

$\cancel{W_h = \sum_{i=I,II} \sum_{\Omega_h} F(d_0^+ u, d_1^+ \ldots)}$

$$W_h = \sum_{i=I,II} \sum_{\Omega_h} F(d_{0,i}^+ u, d_{1,i}^+ u, d_{2,i}^+ u) \Rightarrow$$

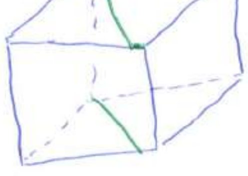
$$= \sum_{i=I,II} \left(d_{0,i}^- \frac{\partial F}{\partial d_{0,i}^+ u} - d_{1,i}^- \frac{\partial F}{\partial d_{1,i}^- u} - d_{2,i}^- \frac{\partial F}{\partial d_{2,i}^- u} \right) \Delta S = 0$$

3D:



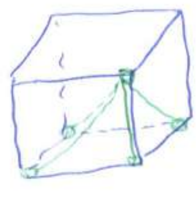
← 5 тетр.

$$\Delta V_1 = \frac{1}{3}\Delta V, \quad \Delta V_2 = \Delta V_3 = \Delta V_4 = \Delta V_5 = \frac{1}{6}\Delta V$$



← можно разбить элемент на 2 призма
и каждая призма на 3 тетраэдра

$$\Delta V_i = \frac{1}{6}\Delta V$$



пусть в ячейке m элементов

~~$\Delta V_m = \gamma_m \Delta V \quad m = 1, \ldots, ?$~~

~~$W_h = \sum_{\ell=1}$~~ $\Delta V_p = \gamma_p \Delta V \qquad p = 1, \ldots, m$

$$W_h = \sum_{p=1}^{m} \gamma_p \sum_{\Omega_h} F\left(d_{0,p}^{+} u,\ d_{1,p}^{+} u,\ d_{2,p}^{+} u,\ d_{3,p}^{+} u\right) \Delta V$$ – сеточный функционал

$$\sum_{p=1}^{m} \gamma_p \left(d_{0,p}^{-} u \frac{\partial F}{\partial d_{0,p}^{+} u} - d_{1,p}^{-} \frac{\partial F}{\partial d_{1,p}^{+} u} - d_{2,p}^{-} \frac{\partial F}{\partial d_{2,p}^{+} u} - d_{3,p}^{-} \frac{\partial F}{\partial d_{3,p}^{+} u} \right) = 0$$

$$\delta W = \int_V \sigma_{ij} \, \delta \varepsilon_{ij} \, dV = \int_V (\rho F - \rho \ddot{u}_i) \, \delta u_i \, dV + \int_S P_i \, \delta u_i \, dS \quad \ldots (1)$$

$$\delta W = 0$$

тензор симметричный $\rightarrow$ $\varepsilon_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i})$

тензор малого поворота $\longrightarrow$ $W_{ij} = \frac{1}{2}(u_{i,j} - u_{j,i})$

$$u_{i,j} = \varepsilon_{ij} + W_{ij}$$

$$\sigma_{ij} \, \delta W_{ij} = 0 \Rightarrow \sigma_{ij} \, \delta \varepsilon_{ij} = \sigma_{ij} \, \delta u_{i,j}$$

можно оставить это в ур-е (1)

$$\int_V \sigma_{ij} \, \delta \varepsilon_{ij} \, dV = \int_V \sigma_{ij} \, \delta u_{i,j} \, dV = \int_S \sigma_{ij} \, u_j \, \delta u_i \, dS - \int_V \sigma_{ij,j} \, \delta u_i \, dV$$

$\ldots (2)$

ур-е (2)

~~можно с~~ приравниваем $\int_S$ из ур-я (1) и (2)

$$\int_S \sigma_{ij} n_j \delta u_i \, dS = \int_S P_i \delta u_i \, dS$$

приравниваем $\int_V$ из ур-я (1) и (2), получим:

$$\int_V (-\sigma_{ij,j} \delta u_i - \rho F_i \delta u_i + \rho \ddot{u}_i \delta u_i) dV = 0$$

$$\Rightarrow \boxed{\sigma_{ij,j} + \rho F_i = \rho \ddot{u}_i}$$

$$\sigma_{ij,j} + \rho F_i = \rho \ddot{u}_i \quad \text{----} \quad (3)$$

$$\sigma_{ij} = \lambda \delta_{ij} \varepsilon_{kk} + 2\mu \varepsilon_{ij} \qquad \varepsilon_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i})$$

Если мы поставим ε_{ij} в ур-е σ_{ij}, то получим:

$$\sigma_{ij} = \lambda \delta_{ij} u_{k,k} + \mu (u_{i,j} + u_{j,i}) \quad \text{-----} \quad (4)$$

поставим ур-е (4) в (3), то получим:

$$\lambda \delta_{ij} u_{k,kj} + \mu (u_{i,jj} + u_{j,ij}) + \rho F_i = \rho \ddot{u}_i$$

$$\lambda \underline{u_{j,ji}} + \mu (u_{i,jj} + \underline{u_{j,ij}}) + \rho F_i = \rho \ddot{u}_i$$

$$\boxed{(\lambda + \mu) u_{j,ij} + \mu u_{i,jj} + \rho F_i = \rho \ddot{u}_i}$$ ← Система уравнения Ламме

$$\boxed{(\lambda + \mu) \,\mathrm{grad\,div}\,\bar{u} + \mu \Delta \bar{u} + \rho \bar{F} = \rho \frac{\partial^2 \bar{u}}{\partial t^2}}$$

Δu : оператор Лапласа

$$\varepsilon_{ij} = \frac{1}{2}\left(d_j^+ u_i + d_i^+ u_j\right) \Rightarrow \sigma_{ij} = \lambda \delta_{ij}\, d_k^+ u_k + \mu\left(d_j^+ u_i + d_i^+ u_j\right)$$

$$\boxed{\lambda u_{j,ji} + \mu\left(u_{i,jj} + u_{j,ij}\right) + \rho F_i = \rho \ddot{u}_i} \rightarrow \begin{matrix} j \to i \\ k \to j \end{matrix}$$

$$\lambda d_i^- d_j^+ u_j + \mu\left(d_j^- d_i^+ u_i + d_j d_i^+ u_i\right) + \rho F_i = \rho \ddot{u}_i$$

$$D_{ij} = d_i^+ d_j^- = d_j^- d_i^+$$

$$D_\Delta = D_{11} + D_{22} + D_{33}$$

$$\boxed{\lambda D_{ji} u_j + \mu D_{ij} u_j + \mu D_\Delta u_i + \rho F_i = \rho \frac{\partial^2 u_i}{\partial t^2}}$$

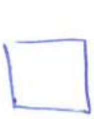
$$(\lambda+\mu)\begin{bmatrix} D_{11}u_1 + D_{12}u_2 + D_{13}u_3 \\ D_{21}u_1 + D_{22}u_2 + D_{23}u_3 \\ D_{31}u_1 + D_{32}u_2 + D_{33}u_3 \end{bmatrix} + \mu D_\Delta \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} + \rho \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix} = \rho D_{tt} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

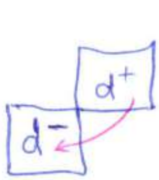
$$\left(D_{tt} f\right)^k = \frac{1}{\tau^2}\left(f^{k+1} - 2f^k + f^{k-1}\right)$$

$u_3 = 0$, $\underbrace{u_1, u_2}_{\text{не зависит от } x_3 \text{ или } u_3}$ — функция (x^1, x^2)

$$(\lambda+\mu)\left(D_{11}u_1 + D_{12}u_2\right) + \mu\left(D_{11}u_1 + D_{22}u_1\right) + \rho F_1 = \rho D_{tt} u_1$$

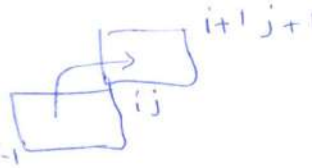
$$(\lambda+\mu)\left(D_{21}u_2 + D_{22}u_2\right) + \mu\left(D_{11}u_2 + D_{22}u_2\right) + \rho F_2 = \rho D_{tt} u_2$$

 $\rightarrow D_{ij} = D_{ji}$

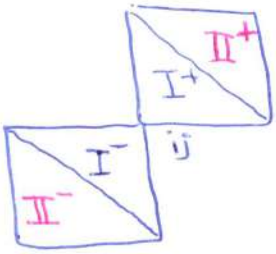


$d_k^+ = T_{11} d_k^-$

↑ оператор сдвига



$$D_{ij} = d_i^+ d_j^- = T_{11}\, d_i^{-*} d_j^- = \underbrace{T_{11}\, d_j^-}_{}\, d_i^- = d_j^+ d_i^- = D_{ji}$$



$\cancel{d^+_{1,I} = T_{11}\, d_{1,II}}$ $\quad d^-_{1,II} = T_{-1-1}\, d^+_{1,I}$ $\qquad d^-_{2,II} = T_{-1-1}\, d^+_{2,I}$

$d^+_{1,I} = T_{11}\, d^-_{1,II} \Rightarrow$ $\qquad d^+_{2,I} = T_{11}\, d^-_{2,II} \Rightarrow$ $\qquad \boxed{d^+_{2,II} = T_{11}\, d^-_{2,I}}$

$\boxed{d^+_{1,II} = T_{11}\, d^-_{1,I}}$

минус

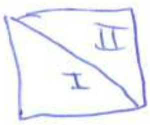
$$D^I_{ij} = d^+_{i,I}\, d^-_{j,I}\ , \qquad d^{II}_{ij} = d^+_{i,II}\, d^-_{j,II}$$

$$= \cancel{T_{11}}\, d^-_{i,I}\, \cancel{T_{-1-1}}\, d^+_{j,I}$$

$$= d^+_{j,I}\, d_{i,I}$$

$$= D^I_{ji}$$

$$D^{II}_{ij} = D^I_{ji}$$



$$D_{ij} \to D^*_{ij} = \frac{1}{2}\left(D^I_{ij} + D^I_{ji}\right)$$

$$D_\Delta =$$

$$\begin{cases} (\lambda+\mu)\left(D^*_{11} u_1 + D^*_{12} u_2\right) + \mu D_\Delta u_1 + \rho F_1 = \rho D_{tt} u_1 \\ (\lambda+\mu)\left(D^*_{12} u_1 + D^*_{22} u_2\right) + \mu D_\Delta u_2 + \rho F_2 = \rho D_{tt} u_2 \end{cases}$$